

Project Title

MedVision AI: Intelligent Medical Image Diagnosis and AI Medical Assistant

Problem Statement:

Doctors often spend significant time analyzing medical images like X-rays, MRI, CT scans, and skin lesion images to detect diseases. In many areas, especially remote locations, access to specialists is limited, and patients find medical reports difficult to understand. **MedVision AI** is an AI-powered assistant that analyzes medical images, predicts possible diseases, and generates simple, patient-friendly explanations to support doctors in faster and more accessible healthcare.

Objective

The project aims to:

- Detect diseases from medical images using **Deep Learning**.
- Provide confidence scores for predictions.
- Explain the prediction in simple language using **Generative AI**.
- Generate an automatic medical summary/report.
- Allow users to ask questions about the prediction through an AI assistant.

Technologies Used

Deep Learning:-

- TensorFlow or PyTorch
- CNN models (EfficientNet, ResNet, DenseNet)

Generative AI:-

- Google Gemini API or OpenAI GPT
- LangChain (optional)
- FAISS (optional for RAG)

Backend:-

- Python
- Flask or FastAPI

Frontend:-

- HTML
- CSS
- Bootstrap
- JavaScript

Database:-

- SQLite or MySQL

Expected Output

- Disease prediction from uploaded medical images.
- Confidence score for the prediction.
- Heatmap showing the important image regions.
- AI-generated explanation in simple language.
- Downloadable medical report (PDF).
- Interactive AI chatbot for user questions.